

# Jacob Kavanal

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## Education

University of California - San Diego *BS, Data Science*

*Sept 2024 - March 2027 — GPA: 3.83*

San Jose State University *BS, Computer Science*

*Aug 2022 - May 2024 — GPA: 3.89*

## Experience

**Data Engineering Intern** *ORiON*

*Danville, CA — Sept 2025 - Present*

- Designed and implemented **ETL pipeline** that handles **~25k job listings/day** using **Python** and **Playwright**, preparing data for downstream ML applications.
- Deployed scalable cloud infrastructures onto **AWS Lambda** with **API Gateway**, utilizing established design patterns to automate data capture and persist results in a **MongoDB** database.
- Utilized **GPT-4o mini** to create vector embeddings for job listings for use downstream

**Co-Founder** *Cove Learning*

*San Diego, CA — Jan 2026 - Present*

- Co-founded a tutoring service and led all technical infrastructure, designing and deploying a full-stack platform from scratch using **TypeScript**, **Supabase**, and **Cloudflare**
- Architected a **PostgreSQL** relational schema to manage tutor-student relationships, session logging, and customer data
- Built and deployed serverless Edge Functions (**Deno**) to handle inbound lead capture from tutor applicants and parent inquiries, with **CAPTCHA verification**, spam filtering, and duplicate prevention
- Implemented tutor portal with **Supabase Auth**, enabling tutors to log sessions, view assigned students, and track hours

**Market Manager & Software Instructor** *CodeAdvantage*

*San Diego, CA — Dec 2023 - Present*

- Rapidly promoted (Instructor → Tech Manager → Market Manager) to direct technical operations, client relations, and curriculum delivery for the San Diego market.
- Standardized curricula for **Python** and **JavaScript**, breaking down complex CS fundamentals into scalable training modules.
- Managed software licensing and hardware infrastructure for **30+ partner schools**, establishing technical SOPs that ensured 100% uptime for daily coding classes.
- Led the recruitment and technical training of new instructors, ensuring consistent teaching quality and adherence to engineering best practices across the region.

**Web Design Intern** *San Jose Works*

*San Jose, CA — Nov 2021 - Dec 2022*

- Led team of three to develop website for the San Jose Works Department, ensuring strict adherence to design specifications.
- Wrote **custom scripts** in **JS** to automate Excel processing of **1,200** applications, reducing analysis time by **75%**.
- Met and presented technical work to representatives of **Mayor Sam Liccardo**.

## Projects

**Mixify** 

*January 2026*

- Developed a **feature engineering pipeline** to analyze **real-time audio signals** (tempo, key, loudness) via **Spotify Web API** for a recommendation system.
- Built a custom algorithm for **signal compatibility** using BPM ( $\pm 5$  range) and harmonics to automate track selection.
- Implemented **OAuth 2.0** with **PKCE flow** for secure authentication, ensuring robust integration between external APIs and the React frontend.

**Recipe Data Analysis**

*June 2025*

- Performed **Exploratory Data Analysis (EDA)** and **Statistical Modeling** using **NumPy** and **pandas** to uncover patterns in seasonal nutrition data.
- Visualized complex dataset trends using **Seaborn** and **matplotlib**, generating insights for further model development.
- Trained and evaluated models with **5-fold CV**, achieving  **$R^2=0.78$**  and **MAE=42.6 kcal** with inference time **12 ms/sample**.
- Assessed model fairness with custom hypothesis testing workflows implemented in Python using **SciPy** and **scikit-learn**.

**Tweet Sentiment Analysis**

*December 2023*

- Built a **Multinomial Naive Bayes Model** using **scikit-learn**, achieving **96.5% accuracy** on a dataset of **2,000 tweets**.
- Implemented an advanced **NLP preprocessing pipeline** (tokenization, negation handling, TF-IDF vectorization) to clean unstructured text data.
- Containerized the application for deployment, demonstrating familiarity with software packaging and scalable architecture.
- Developed a real-time web interface using **React + Node.js** to visualize sentiment classification results.

**Organizations:** SJSU Software Engineering Society (AI/ML Development Team), Data Science Students Society (Outreach)

**Technologies:** Python, Pandas, NumPy, PyTorch, Scikit-learn, **AWS**, Git, Signal Processing, SQL, React, Node.js, MongoDB

**Coursework:** Linear Algebra, Probability & Statistics, Principles of Data Science, Intro to Machine Learning, Data Structures and Algorithms, Computer Architecture, Theoretical Foundations of Data Science